

JUNMING CAO

Senior Engineer | Coding Agents & Post-Training

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Senior Engineer at Huawei with a PhD in Computer Science from Fudan University.

- Built **shared execution and verification infrastructure** for coding agents in accelerator-kernel development, with experience spanning task generation, model evaluation and supervised fine-tuning.
- Currently leading works on **human-agent auditing methods for scalable validation of environments, task specifications and verifiers** used in coding-agent post-training.

Professional Experience

Huawei | Senior Engineer

Sep 2025 – Present

Applied Software Engineering Lab, 2012 Lab

Shanghai, China

■ Coding-Agent Post-Training Data & Evaluation

Aug 2026 – Present

- Leading validation of training tasks and environments reconstructed from **real user-agent traces**, and developing **human-agent auditing methods** for scalable validation of environments, task specifications and verifiers.
- Designed and built a **ScaleSWE-based post-training data pipeline** covering containerised task environments, task and test generation, independent validation, rollout collection with AweAgent and export for supervised fine-tuning (SFT); validated concurrent launch of 256 task instances.
- Fine-tuned Qwen3-30B-A3B-Instruct on 2,954 agent trajectories generated by DeepSeek-v3.2. Improved **SWE-bench Verified from 18.0% to 54.6%** under the same AweAgent evaluation setup.

■ Agent Harness for Ascend NPU Kernel Engineering

Feb 2026 – Present

- Led the architecture and core development of **shared execution and verification infrastructure** across three Ascend NPU kernel agent workflows: kernel generation, numerical debugging and cross-generation migration.
- **Workflow Engine:** 1) Designed a state-machine orchestration engine with explicit stage transitions, retries and termination conditions to address workflow drift, skipped required steps and premature completion in long-running agent tasks. 2) Built a two-layer Hook-Gate mechanism. Hooks enforced workflow constraints during execution, while stage gates independently validated artefacts before allowing stage progression.
- **Results:** Contributed to the team's **#1 ranking on CannBench 910**, achieving a **98.2% correctness pass rate** and **1.15× mean speedup** over expert-written kernels.

■ GPU/NPU Training Consistency

Sep 2025 – Jan 2026

- Diagnosed and resolved **20+ numerical-consistency issues** in LLaMA-Factory through controlled GPU/NPU comparisons and layerwise tensor analysis. Root causes included **model implementation defects, inconsistent data loading and shuffling, and mismatched training hyperparameters**.
- Built automated CI regression tests for training-data handling and loss computation to prevent numerical regressions across training backends.

Selected Research & Publications

- 1 **Exploring the Potential of ChatGPT in Automated Code Refinement: An Empirical Study.** ICSE 2024 (CORE A*, CCF A), co-first author.
Built CodeReview-New, a **14,568-example benchmark for automated code refinement** from review comments. Evaluated ChatGPT against the CodeReviewer model and analysed 400 cases to understand model behaviour and failure modes in review-guided code refinement. Major failure sources included **missing domain knowledge and ambiguous edit locations or review intent**.
- 2 **Understanding Performance Problems in Deep Learning Systems.** ESEC/FSE 2022, first author.
Studied 224 performance problems and built a 58-case reproducible benchmark. Developed **DeepPerf**, a rule-based static checker; evaluation on **1,108 open-source repositories** identified **488 new issues**, of which **27 were fixed**.
- 3 **Understanding Codebase like a Professional! Human-AI Collaboration for Code Comprehension.** ICPC 2026 (CORE A, CCF B), co-author; **ACM SIGSOFT Distinguished Paper Award**.
Co-authored a study of how professional developers understand unfamiliar codebases, based on interviews with 8 professional code auditors, and translated the findings into **CodeMap, an LLM-powered system with hierarchical codebase visualisations and interactive navigation**. Among experienced developers, CodeMap reduced time spent reading LLM responses by **79%**.
- 4 **RegTrieve: Reducing System-Level Regression Errors for Machine Learning Systems via Retrieval-Enhanced Ensemble.** FSE 2025 (CORE A*, CCF A), first author.
Developed RegTrieve, a retrieval-enhanced ensemble that dynamically combines old and updated models to reduce **system-level regression errors** through multi-model ML systems.
- 5 **Understanding Performance Problems in CUDA Programs.** FSE 2026, second author.

Open Source Contributions

- **Committer, canbot community:** Contributed Ascend kernel-migration orchestration engine and layered knowledge management through merged contributions.
- Contributed the GPU/NPU training-diagnostics workflow as an open-source **MindStudio Agent skill**, using msProbe-based layerwise tensor comparison.
- Contributed data-pipeline parallelisation fixes derived from DeepPerf findings; merged into **Keras #15295** and **TensorFlow Agents #650**.

Education & Internships

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| ■ Fudan University PhD, Computer Science; advised by Bihuan Chen and Xin Peng. | 2021 – 2025 |
| Visiting PhD Student, Singapore Management University, Sep 2022 – Mar 2023. | |
| ■ Shanghai Jiao Tong University MSc, Software Engineering | 2018 – 2021 |
| ■ Shanghai Jiao Tong University BSc, Software Engineering | 2014 – 2018 |
| ■ Internships | 2017 – 2020 |
| 1) ByteDance: advertising backend (2020); 2) Momenta: HD-map platform (2019–2020); 3) Tencent Youtu AI Lab: training data collection and model serving (2017). | |

Technical Skills

AI & ML: LLM Post-Training, Training Data Generation and Evaluation.

Coding Agents: Harness Engineering, Executable Task Environments.

Languages & Tools: Python, C++, Docker, Git, AWS, CI/CD.